



**SRM Institute of Science and Technology School
of Computing
Department of Computing Technologies**

**21DCS201P DESIGN THINKING &
METHODOLOGY
REPORT**

Name of the Student : D Bhanu prasad

Register Number : RA2411003011339

Section : A2

Class : II B.Tech CSE (CORE)

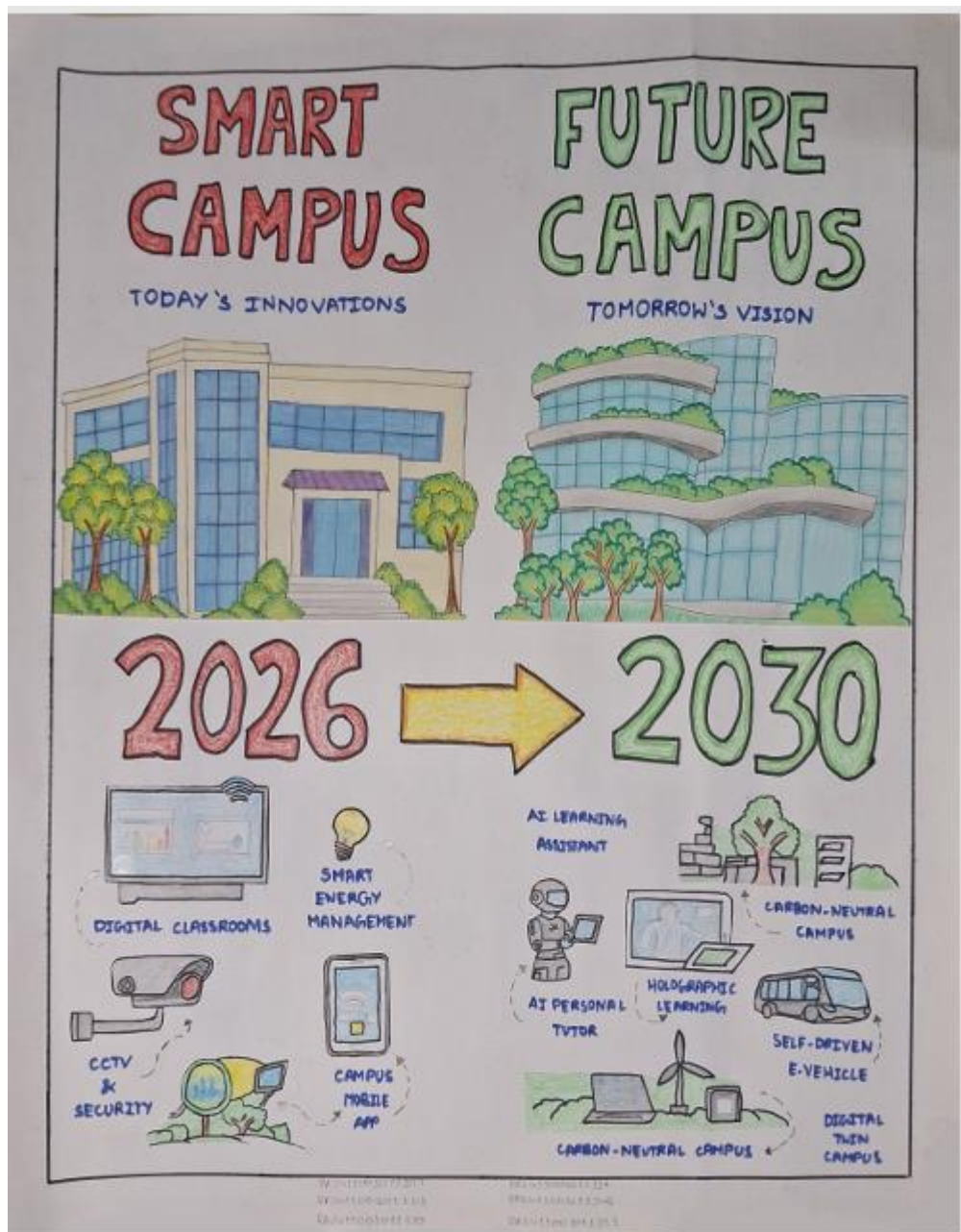
Academic Year / Sem : 2025-2026 (EVEN) / IV SEM

S.No	Activity
1	Visualization
2	Brainstorming
3	Mind Mapping
4	30 Circles
5	Problem Sizing
6	Persona
7	Convergent & Divergent Thinking
8	Business Model Canvas

Faculty In-charge

Dr. M.Kandan, Assistant Professor/CTECH

VISUALIZATION



BRAINSTROMING

THE CREDIT ACCESS GAP

PROBLEM STATEMENT

Many people like gig workers (swiggy / zomato drivers), small shop owners, small businesses, and daily wage workers cannot get loans from banks. So these people are forced to borrow from private lenders.

PROBLEM

Banks only lend to people who already have a loan history, but how do get a history of loan. If no one will give you a first loan. Millions who pay bills on time still get rejected because banks ignore that data. 7 in 10 first time applicants are turned down.

CONTEXT

Everytime someone applies for a loan to buy stock for their shop, cover a medical bill or grow a small business, they get rejected in seconds with no reason given.

ALTERNATIVE

- * Apply to a bank - rejected due to no loan history.
- * Visit a branch and fill a lot of paper work
- * Ask family or friends for money.
- * Borrow from a local money lender at a very high interest.

USERS

- * Delivery and rickshaw drivers
- * Freelancers and self-employed workers.
- * Small shop and Kirana store owners.
- * Small business owners.
- * Daily wage workers who pay bills digitally.
- * Students and Fresh graduates.

EMOTIONAL IMPACT

Embarrassed and frustrated by unexplained rejections. Worried about their business and frustrated that

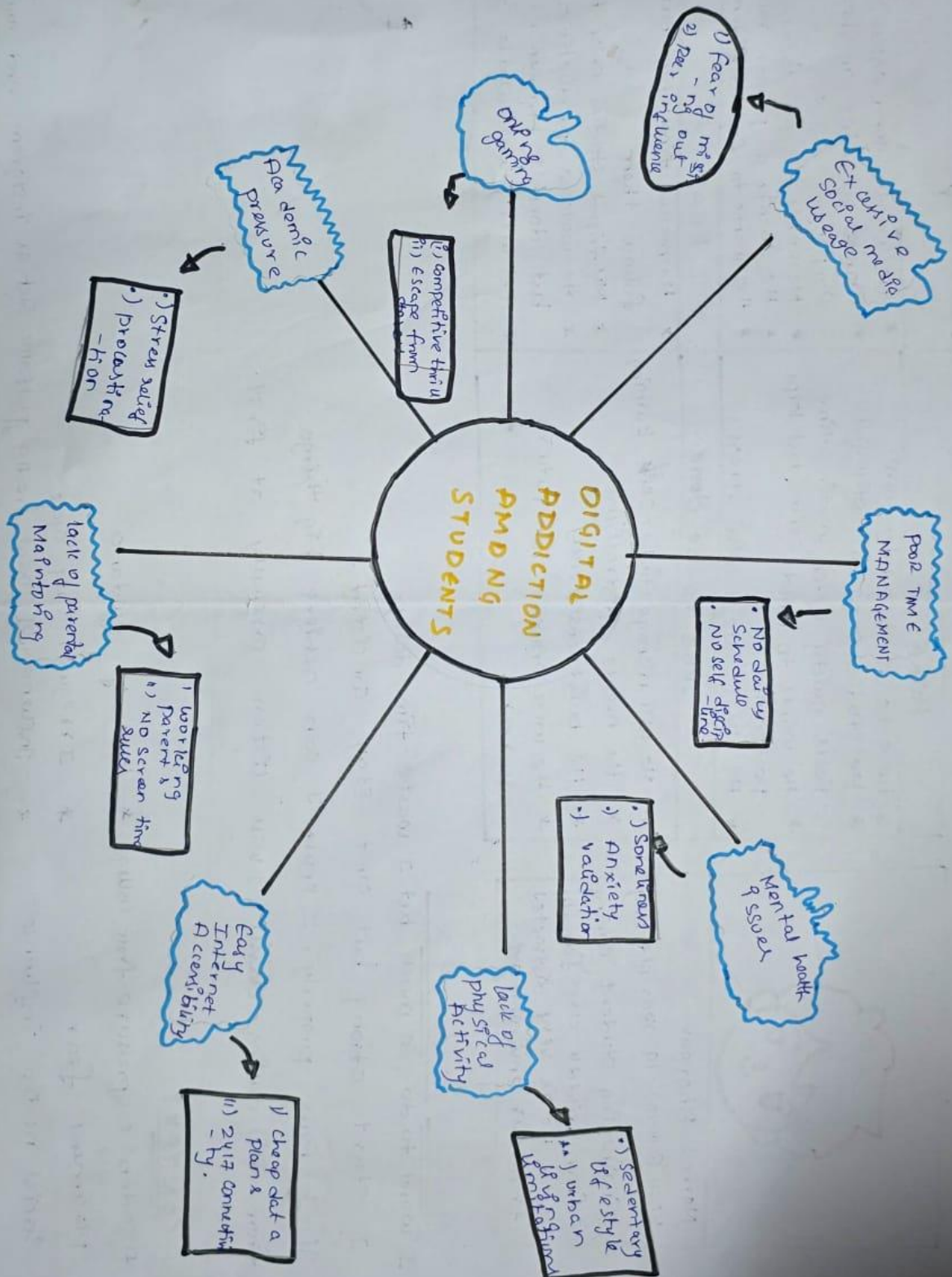
QUANTIFIABLE IMPACT

- * 1.4 billion people worldwide have no access to loans.
- * 95 million small businesses in India are shut down.
- * Loan approval takes 7-10 days.

SHORTCOMINGS

- * Banks: Won't lend without a prior loan.
- * deadend for first timers.
- * Branch Visit: Slow, lots of paper work takes over a week.
- * Family - Friends: Strains relationships. limited amounts.
- * None of these look at whether you actually pay your bills on time.

MIND MAPPING



30 CIRCLES



PROBLEM SIZING

1. PROBLEM DIGITAL PAYMENTS FAILURES FOR SMALL SHOP OWNERS

11:05

D. Bhanu prasad
R22411003011339



5. evolution: business

- * Growth of UPI and digital payments
- * Government push for cashless economy
- * Increase in smart phone users

2. User Segment

- * Small Shop owners
- * Kirana Store owners
- * Small retail businesses
- * Around 12-15 million small shops in India.

3. Frequency

- * Payment failure happens 2-3 times per week.
- * Around 8-12 times per month
- * Approximately 100-150 times per year.

4. Severity

- * ₹100-₹500 loss or delay per failed transaction
- * Time spent resolving ₹500.
- * Estimated yearly loss: ₹10L - ₹30L per shop for ex.

6. evolution: societal

- * Better banking service technology.
- * Improved payment instruction
- * faster refund system.
- * stronger cybersecurity systems.

Annual Problem size ≈ ₹432 Billion per year
(₹43,200 crore per year)

PERSONA

R22U11003011, 33 9
D. Bharu Prasad



Name: Bhargav
Bach Ground, 19 year old boy
Engineering student coming
from middle class family
and he is very distracted
from his environment.

STORIES / OBSERVATIONS

- * I want to do so much but I waste time daily.
- * I start strong but can't stay consistent.
- * If I focus properly, I know I can achieve big things
- * from now during class, I will listen carefully at first.

CONTEXT

- * Academic pressure from college.
- * Placement fear
- * Social media influence
- * Lack of guidance
- * Irregular sleep cycle
- * Influence of social media platform such as Instagram and youtube.

Pains

- * He gets distracted by phone.
- * He studies a lot but still feels.
- * feels guilty after wasting time
- * He wants to build stress but keeps postponing.
- * He sleeps late due to stress.

Jobs to be done

- * He will manage study & self learning
- * He avoids distraction
- * He build consistency
- * He will prepare for future career.

Gains

- * He wants good CGPA (9+)
- * wants placement in high-paying company.
- * He wants to stay focused daily.
- * He wants a proper routine
- * He wants to feel confident before exam

REALITY

- * He makes promises but doesn't follow them
- * My friend studies only before exam.
- * He watches productivity videos but doesn't implement
- * He uses phone while studying

Convergent & Divergent Thinking

CLASSROOM ATTENDANCE SYSTEM				
RA2411003011321	RA2411003011339	RA2411003011346	RA2411003011351	RA2411003011355
1. Manual attendance register.	> Online Attendance Portal for Teachers.	> Digital Attendance using laptops in class.	> Cloud-based Attendance management System.	> Automated Attendance Generator.
2. Student ID CARD Scanning.	> Tablet Based Attendance System.	> Teacher dashboard for Quick Attendance making.	> Attendance Alert System for parents.	> Attendance track integrated with LMS.
3. QR Code attendance System.	> GPS-based attendance through mobile APP.	> Automated Attendance through classroom Sensors.	> Smart gate entry Attendance System.	> Smart ID cards & automatically reg Attendance.
4. Mobile APP Attendance System.	> Smart classroom Camera Detecting camera.	> AI-Based System Identifying students in class.	> Attendance through classroom wifi connection detection.	> AI System that count students using a classroom camera.
5. Biometric fingerprint Attendance.	> Voice recognition Attendance System for Student ID card.	> Attendance Confirmation through OTP.	> Voice Command System where students say their ID number using voice recognition.	> Voice Recognition checkin.
6. face recognition attendance System.	> Barcode Scanning.	> Group Attendance System for lab.	> Eye Scan biometric Attendance System.	> Seat pressure Sensor.
7. RFID Card Attendance System.	> Bluetooth based Automatic Attendance.	> Smart watch Attendance Check in.	> Attendance through classroom entry gate.	> Location Based Attendance.
8. NFC Enabled Attendance System.	> Attendance through College Wifi Login.	> Attendance making through fingerprint on phone.	> Automatic Attendance through smart watch checkin.	> Attendance through Code Scanning at the classroom.

STEP-2 (RA2411003011339)		STEP-3		STEP-4	
- AI eye-contact detection system - marks attendance only when student looks at board. - AI neural behaviour mapping - tracks cognitive engagement patterns. - IoT smart shoe detection system - identifies student via shoe sensors. - AI emotion recognition attendance - confirms presence via emotional response. - IoT smart calling grid - tracks each student's position in real-time. - AI learning pattern recognition.		- The class turned into a brain storming hub full of crazy ideas. - Our class design a futuristic attendance system using AI. - The class felt like a startup meeting during the activity. - Every student in the class contributed a unique solution. - Class we design thinking to solve real-world problems.		YES BUT... - Yes, but face recognition may fail if lighting is poor. - Yes, but RFID cards can be misused by friends (proxy). - Yes, but mobile based attendance depends on internet. YES AND... - Yes, AND we can add infrared cameras to improve accuracy. - Yes, AND we can combine it with face recognition for verification. - Yes, AND we can store data offline and sync later.	
STEP-2 (RA2411003011346)		STEP-3		STEP-4	
- IoT heat-zone mapping system - maps unique temperature zones per student. - IoT smart light reflection system - detects body reflection patterns. - AI typing rhythm recognition - detects student via typing speed/pattern. - IoT micro-location beacons - tracks exact seat/row position. - AI lip-movement recognition - identifies students when they speak.		- The room turned into an innovation lab during activity. - I don't like the room as every one should brainstorming. - I entered the room early in the morning to prepare for class. - All the students gathered in the room before the teacher arrived. - The teacher explained the lesson only in the room.		YES BUT... - Yes, but GPS-based attendance may not be accurate indoors. - Yes, but biometric system takes time for large classes. - Yes, but AI camera system are expensive. YES AND... - Yes, AND we can use wifi or bluetooth for better indoor tracking. - Yes, AND we can use multiple cameras to reduce waiting time. - Yes, AND we can use one shared system for multiple classrooms.	
STEP-2 (RA2411003011351)		STEP-3		STEP-4	
- IoT smart floor tiles - detects pressure patterns/identity. - AI classroom attention heatmap - marks attendance based on focus. - IoT classroom energy usage tracking - maps device usage to student. - AI classroom shadow mapping - identifies students using dynamic shadows. - AI interaction-based attendance - marks attendance when student interaction/inclusion. - AI neural behaviour mapping.		- There are many tasks and chat arranged neatly inside the hall. - Energy in the hall increased when creative ideas were shared. - Our hall design thinking to solve real-world problems. - The hall transformed into simple problem into a innovative project.		YES BUT... - Yes, but students may try to cheat the system. - Yes, but wearable device can be forgotten at home. - Yes, but continuous tracking may raise privacy concerns. YES AND... - Yes, AND we can use AI to detect unusual patterns. - Yes, AND we can add a backup manual system. - Yes, AND we can store only attendance data and not personal data.	

RA2411003011339

Ideas	321	346	351	352	339	Total
* online Attendance portal for teachers	100	100	100	10	-	310
* Tablet-Based Attendance portal for Students.						
* GPS-based attendance through mobile App.	1000	1000	1000	100	-	3100
* Smart classroom Camera detecting camera.	1000	100	1000	1000	-	3100
* Voice recognition Attendance System for Students.	1000	1000	1000	1000	-	4000
* Bluetooth based Automatic attendance	100	10	100	1000	-	1230
* Attendance through College WiFi login						
* AI eye-contact detection System - marks attendance only when student looks at board	100	100	100	10	-	310
* AI neural behaviour mapping - tracks cognitive engagement patterns.	1000	100	100	100	-	1300
* IoT Smart shoe detection System - identifies Student via shoe sensors.	100	10	10	100	-	220
* AI emotion recognition attendance - confirms presence via emotional response.	100	100	100	100	-	400
* IoT Smart calling grid - tracks each student's position in real-time.	1000	1000	100	1000	-	3100
* The class turned into a brainstorming hub full of crazy ideas.	100	100	100	1000	-	1300
* Our class designed a futuristic attendance system using AI.	100	1000	100	1000	-	2200
* The class felt like a startup meeting during the activity.	100	100	100	100	-	400
* Every student in the class contributed a unique solution.	1000	1000	100	100	-	2200
* class use design thinking to solve real-world problems.	1000	1000	1000	1000	-	4000
* AI learning pattern recognition.	100	100	1000	1000	-	2200

Business Model Canvas

